

ReliCheck Plug-In Analysis Apps

Tier 1: Overview Apps

These are used in every studio because every project needs a basic evidence snapshot.

Plug-In App	What It Does	Studios
Project Snapshot App	Shows project purpose, data sources, sample size, response count, completion status	All
Sample Profile App	Summarizes who is in the dataset by group, demographic, rater type, class, role, etc.	All
Data Readiness App	Checks whether the project has enough data and structure to begin analysis	All
Missing Data App	Shows missingness by item, variable, group, respondent, or section	All
Data Quality App	Flags duplicates, invalid values, straight-lining, incomplete records, outliers, and low-effort responses	All

These should be foundational. They tell the user: *What do I have, and is it ready?*

Tier 2: Instrument Quality Apps

These are central to ReliCheck's identity because they evaluate whether the tool itself is strong enough.

Plug-In App	What It Does	Studios
ReliCheck Strength Index App	Composite quality score for the instrument/evidence system	Survey, MM, TIA, 360
Reliability App	Cronbach's alpha, McDonald's omega, item-rest correlations, alpha if removed	Survey, MM, TIA, 360
Validity Review App	Reviews construct alignment, content validity, face validity, and interpretation risk	Survey, MM, TIA, 360
Item Quality App	Flags unclear, double-barreled, biased, leading, vague, or misaligned items	Survey, TIA, 360, MM
Scale Structure App	Groups items into scales and checks whether the scale structure makes sense	Survey, TIA, 360
Factor Readiness App	KMO, Bartlett's test, sample adequacy, and whether factor analysis is appropriate	Survey, MM, TIA
Factor Analysis App	Exploratory factor analysis, factor loadings, cross-loadings, dimensionality	Survey, MM, TIA
Response Scale Review App	Reviews Likert scales, rating ranges, anchors, midpoint use, and response-option balance	Survey, TIA, 360

Plug-In App	What It Does	Studios
Bias and Clarity Review App	AI-assisted review of item wording, cultural assumptions, accessibility, and interpretation risk	Survey, TIA, 360

This is where ReliCheck is different from SurveyMonkey. It should not just analyze responses. It should evaluate the evidence tool.

Tier 3: Descriptive Analysis Apps

These should be shared across almost all studios.

Plug-In App	What It Does	Studios
Descriptive Statistics App	Mean, median, mode, SD, min, max, range, variance	All
Frequency App	Counts and percentages for categorical variables and survey items	All
Cross-Tabs App	Breaks down responses by group, role, demographic, rater type, etc.	All
Distribution App	Histograms, bar charts, density summaries, skewness, kurtosis	Survey, MM, TIA, 360
Group Summary App	Means, percentages, and counts by group	All
Top/Bottom Item App	Shows strongest and weakest items, competencies, outcomes, or themes	Survey, TIA, 360
Scale Score App	Computes composite scores from item groups	Survey, MM, TIA, 360
Theme Count App	Counts qualitative theme frequency and prevalence	MM, Survey, TIA, 360
Open-Ended Summary App	Summarizes open-ended response patterns	MM, Survey, TIA, 360

These apps support the question: *What does the data show before we test anything?*

Tier 4: Inferential Analysis Apps

These are the statistical testing plug-ins.

Plug-In App	What It Does	Studios
T-Test App	Compares means between two groups	Survey, MM, TIA, 360
Welch T-Test App	Two-group mean comparison when variances are unequal	Survey, MM, TIA, 360
Paired T-Test App	Compares paired/pre-post/self-other scores	Survey, MM, TIA, 360
ANOVA App	Compares means across three or more groups	Survey, MM, TIA, 360
Welch ANOVA App	ANOVA when homogeneity of variance is violated	Survey, MM, TIA, 360
Post-Hoc Comparison App	Tukey, Games-Howell, pairwise comparisons after ANOVA	Survey, MM, TIA, 360
Chi-Square App	Tests relationships between categorical variables	Survey, MM, TIA, 360
Correlation App	Pearson, Spearman, point-biserial where appropriate	Survey, MM, TIA, 360
Regression App	Linear/logistic regression for prediction and association	Survey, MM, TIA
Effect Size App	Cohen's d, eta squared, omega squared, Cramer's V, odds ratio	All
Confidence Interval App	Adds uncertainty ranges around estimates and differences	All
Assumption Check App	Normality, variance equality, independence, sample size adequacy	Survey, MM, TIA, 360

These should be modular because the same test may appear in multiple studios with different language.

For example, ANOVA in Survey Studio may compare satisfaction by department. In TIA Studio, it may compare test scores by section. In 360 Studio, it may compare competency ratings by rater group. Same statistical engine, different interpretation.

Tier 5: Interpretation Apps

These convert results into meaning.

Plug-In App	What It Does	Studios
Key Findings App	Pulls the most important findings from all completed analyses	All

Plug-In App	What It Does	Studios
Practical Significance App	Explains whether findings matter in real-world terms, not just statistically	All
Limitations App	Flags sample size, missing data, weak reliability, design limits, and overclaim risks	All
Recommended Actions App	Converts findings into next steps	All
AI Interpretation App	Generates plain-language explanations from computed results	All
Teaching Moments App	Explains methods to the user in plain language	All
Decision Readiness App	Says whether the evidence is strong enough for exploratory, operational, or high-stakes use	All
Evidence Alignment App	Checks whether results support, complicate, or contradict the original purpose	All

This layer is one of ReliCheck’s strongest differentiators. It makes analysis understandable without dumbing it down.

Tier 6: Reporting Apps

These should also be plug-ins.

Plug-In App	What It Does	Studios
Report Builder App	Builds the final report from selected analysis modules	All
Executive Summary App	Creates a plain-language summary for decision-makers	All
Methodology Section App	Writes method, sample, variables, and analysis approach	All
Findings Section App	Converts results into report-ready findings	All
Tables and Figures App	Creates clean tables, charts, and appendix-ready outputs	All
Recommendations App	Generates recommended actions based on findings and limitations	All
Export App	PDF, Word, CSV, Excel, PowerPoint-ready summaries	All
Appendix App	Adds technical tables, item diagnostics, assumptions, and detailed outputs	All

Reports should be modular. Each analysis app should be able to contribute a report section.

Studio-Specific Plug-In Apps

Some apps are not common to all studios, but they are still important.

Survey Studio Specific

Plug-In App	Purpose
Survey Strength Index App	Scores survey quality and trustworthiness
Question Wording Review App	Reviews survey item wording
Scale Reliability App	Alpha, omega, item-rest, alpha if removed
Response Pattern App	Straight-lining, acquiescence, speeders, nonresponse
Survey Revision App	Suggests improved item wording and structure

MM Studio Specific

Plug-In App	Purpose
Codebook Builder App	Creates and manages qualitative codes
Theme Analysis App	Identifies and summarizes themes
Theme-by-Group App	Compares themes across groups
Quote/Evidence Extractor App	Pulls representative quotes for findings
Joint Display App	Connects quantitative results with qualitative themes
Integration Quality App	Evaluates convergence, complementarity, divergence, and expansion
Qual-to-Quant Variable Builder App	Converts qualitative patterns into structured variables

TIA Studio Specific

Plug-In App	Purpose
Item Difficulty App	Shows percent correct or difficulty level
Item Discrimination App	Shows how well items distinguish high and low performers
Distractor Analysis App	Examines answer choices for multiple-choice items
Answer Key Validation App	Checks possible scoring/key issues
Bloom's Alignment App	Aligns items and rubrics to cognitive demand
Rubric Builder App	Builds analytic or holistic rubrics
Rubric Quality Review App	Reviews clarity, alignment, levels, and scoring fairness

Plug-In App	Purpose
Open-Ended Scoring App	Supports essay/constructed response scoring
Learning Outcome Alignment App	Checks whether items measure intended outcomes

360 Studio Specific

Plug-In App	Purpose
Rater Group Comparison App	Compares self, peer, supervisor, direct report, stakeholder ratings
Self-Other Gap App	Shows perception gaps
Competency Profile App	Summarizes strengths and development areas
Confidentiality Threshold App	Suppresses unsafe small-group results
Comment Theme App	Summarizes open-ended leadership feedback
Development Plan App	Converts feedback into coaching priorities
Cohort Summary App	Aggregates 360 results across a group

Highest-Priority Plug-In Apps to Build First

I would not build everything at once. Start with the reusable core.

Build First

Priority	Plug-In App	Why
1	Data Readiness App	Every studio needs to know whether data can be analyzed
2	Descriptive Statistics App	Common to all studios
3	Missing Data App	Essential before analysis
4	Reliability App	Core to instrument quality
5	ReliCheck Strength Index App	Signature ReliCheck feature
6	Open-Ended Summary App	Used across Survey, MM, TIA, and 360
7	T-Test / ANOVA / Chi-Square / Correlation Apps	Core inferential suite
8	Effect Size App	Makes results more meaningful
9	AI Interpretation App	Turns outputs into understandable findings
10	Report Builder App	Converts analysis into product value

